

To get the first Adem relation from (1) we choose n and ℓ so that the left side of (1) has only one term, namely we take $n = 2(1 + p + \cdots + p^{r-1}) + 2s$ and $\ell = mn + s$ for given integers r and s . Then

$$\binom{m(n-2j)}{mn+j-\ell} = \binom{p^{r-1}-(p-1)(j-s)}{j-s}$$

and if r is large, this binomial coefficient is 1 if $j = s$ and 0 otherwise since if the rightmost nonzero digit in the p -adic expansion of the 'denominator' $j - s$ is x , then the corresponding digit of the 'numerator' $(p-1)[(1+p+\cdots+p^{r-1}) - (j-s)]$ is obtained by reducing $(p-1)(1-x) \bmod p$, giving $x-1$, and $\binom{x-1}{x} = 0$. Then (1) becomes

$$\begin{aligned} (-1)^{i+s} P^i P^s(\iota) &= \sum_j (-1)^{\ell+j} \binom{m(n-2j)}{mn+j-i} P^{i+s-j} P^j(\iota) \\ \text{or } P^i P^s(\iota) &= \sum_j (-1)^{i+j} \binom{m(n-2j)}{mn+j-i} P^{i+s-j} P^j(\iota) \quad \text{since } \ell \equiv s \bmod 2 \\ &= \sum_j (-1)^{i+j} \binom{m(n-2j)}{i-pj} P^{i+s-j} P^j(\iota) \quad \text{since } \binom{x}{y} = \binom{x}{x-y} \\ &= \sum_j (-1)^{i+j} \binom{p^{r+(p-1)(s-j)-1}}{i-pj} P^{i+s-j} P^j(\iota) \end{aligned}$$

If r is large and $i < ps$, the term p^r in the binomial coefficient can be omitted since we may assume $i \geq pj$, hence $j < s$, so $-1 + (p-1)(s-j) \geq 0$ and the p^r has no effect on the binomial coefficient if r is large. This shows the first Adem relation holds for the class ι , and the general case follows as in the case $p = 2$.

To get the second Adem relation we choose $n = 2p^r + 2s$ and $\ell = mn + s$. Reasoning as before, the left side of (2) then reduces to $(-1)^{i+s} P^i \beta P^s(\iota)$ and (2) becomes

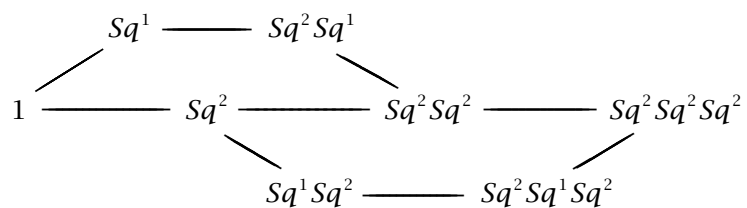
$$\begin{aligned} P^i \beta P^s(\iota) &= \sum_j (-1)^{i+j} \binom{(p-1)(p^r+s-j)}{i-pj} \beta P^{i+s-j} P^j(\iota) \\ &\quad - \sum_j (-1)^{i+j} \binom{(p-1)(p^r+s-j)-1}{i-pj-1} P^{i+s-j} \beta P^j(\iota) \end{aligned}$$

This time the term p^r can be omitted if r is large and $i \leq ps$. □

Exercises

1. Determine all cohomology operations $H^1(X; \mathbb{Z}) \rightarrow H^n(X; \mathbb{Z})$, $H^2(X; \mathbb{Z}) \rightarrow H^n(X; \mathbb{Z})$, and $H^1(X; \mathbb{Z}_p) \rightarrow H^n(X; \mathbb{Z}_p)$ for p prime.
2. Use cohomology operations to show that the spaces $(S^1 \times \mathbb{CP}^\infty)/(S^1 \times \{x_0\})$ and $S^3 \times \mathbb{CP}^\infty$ are not homotopy equivalent.
3. Since there is a fiber bundle $S^2 \rightarrow \mathbb{CP}^5 \rightarrow \mathbb{HP}^2$ by Exercise 35 in §4.2, one might ask whether there is an analogous bundle $S^4 \rightarrow \mathbb{HP}^5 \rightarrow \mathbb{OP}^2$. Use Steenrod powers for the prime 3 to show that such a bundle cannot exist. [The Gysin sequence can be used to determine the map on cohomology induced by the bundle projection $\mathbb{HP}^5 \rightarrow \mathbb{OP}^2$.]
4. Show there is no fiber bundle $S^7 \rightarrow S^{23} \rightarrow \mathbb{OP}^2$. [Compute the cohomology ring of the mapping cone of the projection $S^{23} \rightarrow \mathbb{OP}^2$ via Poincaré duality or the Thom isomorphism.]

5. Show that the subalgebra of $\mathcal{A}_2$ generated by Sq^i for $i \leq 2$ has dimension 8 as a vector space over $\mathbb{Z}_2$, with multiplicative structure encoded in the following diagram, where diagonal lines indicate left-multiplication by Sq^1 and horizontal lines indicate left-multiplication by Sq^2 .



Appendix

Topology of Cell Complexes

Here we collect a number of basic topological facts about CW complexes for convenient reference. A few related facts about manifolds are also proved.

Let us first recall from Chapter 0 that a CW complex is a space X constructed in the following way:

- (1) Start with a discrete set X^0 , the 0-cells of X .
- (2) Inductively, form the n -skeleton X^n from X^{n-1} by attaching n -cells e_α^n via maps $\varphi_\alpha: S^{n-1} \rightarrow X^{n-1}$. This means that X^n is the quotient space of $X^{n-1} \amalg_\alpha D_\alpha^n$ under the identifications $x \sim \varphi_\alpha(x)$ for $x \in \partial D_\alpha^n$. The cell e_α^n is the homeomorphic image of $D_\alpha^n - \partial D_\alpha^n$ under the quotient map.
- (3) $X = \bigcup_n X^n$ with the weak topology: A set $A \subset X$ is open (or closed) iff $A \cap X^n$ is open (or closed) in X^n for each n .

Note that condition (3) is superfluous when X is finite-dimensional, so that $X = X^n$ for some n . For if A is open in $X = X^n$, the definition of the quotient topology on X^n implies that $A \cap X^{n-1}$ is open in X^{n-1} , and then by the same reasoning $A \cap X^{n-2}$ is open in X^{n-2} , and similarly for all the skeleta X^{n-i} .

Each cell e_α^n has its **characteristic map** Φ_α , which is by definition the composition $D_\alpha^n \hookrightarrow X^{n-1} \amalg_\alpha D_\alpha^n \rightarrow X^n \hookrightarrow X$. This is continuous since it is a composition of continuous maps, the inclusion $X^n \hookrightarrow X$ being continuous by (3). The restriction of Φ_α to the interior of D_α^n is a homeomorphism onto e_α^n .

An alternative way to describe the topology on X is to say that a set $A \subset X$ is open (or closed) iff $\Phi_\alpha^{-1}(A)$ is open (or closed) in D_α^n for each characteristic map Φ_α . In one direction this follows from continuity of the Φ_α 's, and in the other direction, suppose $\Phi_\alpha^{-1}(A)$ is open in D_α^n for each Φ_α , and suppose by induction on n that $A \cap X^{n-1}$ is open in X^{n-1} . Then since $\Phi_\alpha^{-1}(A)$ is open in D_α^n for all α , $A \cap X^n$ is open in X^n by the definition of the quotient topology on X^n . Hence by (3), A is open in X .

A consequence of this characterization of the topology on X is that X is a quotient space of $\amalg_{n,\alpha} D_\alpha^n$.